FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Whole Truth

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is a 1/12 block shorter than a 1/3 block if 12 is greater than 3?'
- Ask: 'How does the whole datum frame act as an automatic error checker?'
- Ask: 'Can five 1/4 blocks fit inside the 84.00 mm whole frame? Why not?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 1/4 is bigger than 1/2 because 4 > 2. Direct them to stand the physical blocks side-by-side.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a tower made of three DIFFERENT unit fraction blocks. My summit levels exactly to the whole. None of my blocks"
" is 1/2. What blocks am I?"

OFFICIAL SOLUTION KEY:

Blocks [1/3], [1/4], and [1/12] (since 1/3 + 1/4 + 1/12 = 4/12 + 3/12 + 1/12 = 8/12... actually 1/2+1/3+1/6=1).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Whole Truth

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does a bigger denominator mean a smaller piece? In this lab, you test the physical whole datum and discover inverse proportionality.

STEP-BY-STEP APPARATUS SETUP:

1. Place the Whole Frame [1/1] (84.00 mm tall) vertically on your desk.
2. Stack two [1/2] blocks. Compare their height against the [1/1] whole.
3. Stack four [1/4] blocks beside it. Observe the three summits.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/1], [1/2], [1/3], [1/4], [1/6], [1/12] + Stacking Base

DATUM TARGET: H(1/n) = 84.00 / n mm -> H(1/1)=84.00, H(1/2)=42.00, H(1/12)=7.00 mm

CALCULATOR KEY: [1] [ENTER] [2] [/] -> 0.5

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3A, Chapter 6 (Understanding Fractions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 3, Unit 4 (Equal Shares & Wholes)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 3, Ch. 8 • CCSS.MATH.3.NF.A.1 • TEKS 3.3A
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5 ('Fractions & Ratios'), pp. 98–118
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (Inverse Linear Formula) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE SUMMIT LEVELING CHECK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Lay a straightedge across Tower [1/1], Tower [1/2]+[1/2], and Tower four [1/4]s. All level to exactly 84.00 mm!

Step 2: Pull Block [1/12] (height 7.00 mm). How many [1/12] blocks make a half? (Stack 6 blocks).

Step 3: Try to make a whole with three [1/3] blocks. Confirm they match 84.00 mm.

Step 4: Notice: As denominator n grows (2 -> 3 -> 4 -> 12), block height shrinks (42 -> 28 -> 21 -> 7 mm).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(1/n) = 84.00 / n mm -> H(1/1)=84.00, H(1/2)=42.00, H(1/12)=7.00 mm

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/1], [1/2], [1/3], [1/4], [1/6], [1/12] +
Stacking Base

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MISSING PIECE PUZZLE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

If you stack one [1/2] block and one [1/3] block, what single block is needed to complete the 84.00 mm whole?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure 1/2 + 1/3 gap against the 84.00 mm frame.**
   Expected Physical Finding: **Gap = 14.00 mm**

2. **Which block has height 14.00 mm? Test [1/6].**
   Expected Physical Finding: **Block [1/6] levels the summit!**

3. **Write the fraction equation: 1/2 + 1/3 + ? = 1.**
   Expected Physical Finding: **1/2 + 1/3 + 1/6 = 1**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [ENTER] [2] [/] -> 0.5

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Numerator 1
ENTER	Y: 1 | X: 1	Stack lift
2	Y: 1 | X: 2	Denominator 2
/	X: 0.5	Divide: evaluates fraction value

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Multi-Face Transformer

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does 1/4 turn into 90° when talking about a clock or circle?'
- Ask: 'If you know 1/5 is 20%, what must 2/5 and 3/5 be?'
- Ask: 'How does rotating the block help when solving word problems with money?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 0.25 is larger than 0.5 because 25 > 5. Show them that two 0.25 blocks stacked equal one 0.5 0 block.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"My front face says 1/8. What does my decimal face, percent face, and angle face say?"
""

OFFICIAL SOLUTION KEY:

Decimal = 0.125, Percent = 12.5%, Angle = 45°.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Multi-Face Transformer

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Fractions, decimals, percentages, and circle angles are not four different subjects—they are just four faces of the exact same physical block!

STEP-BY-STEP APPARATUS SETUP:

1. Pick up Block [1/4] in your hand.
2. Look at the Front Face: notice debossed '1/4'.
3. Rotate the block 90° to the right, then 180°, then 270°.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/2], [1/4], [1/5], [1/10] + Desk Comparator Stage

DATUM TARGET: Front: 1/4 | Right: 0.25 | Back: 25% | Left: 90° (Geometric Equivalence)

CALCULATOR KEY: [1] [/] [4] [F-D] -> 0.25

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 4B, Chapter 9 (Fractions & Decimals)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 3 (Decimals & Hundredths)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 7 • CCSS.MATH.4.NF.C.6 • TEKS 4.2G
GRADE LEVEL BAND:	Grades 4–6 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 102–108
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (4-Face Invariant) & Page 5 (Tutorial 1)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE 4-WAY ROTATION AUDIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Read the 4 faces of Block [1/4]: Front='1/4', Right='0.25', Back='25%', Left='90°'.

Step 2: Now inspect Block [1/2]: Front='1/2', Right='0.50', Back='50%', Left='180°'.

Step 3: Stack two [1/4] blocks with angle faces showing: 90° + 90° = 180°! It matches Block [1/2].

Step 4: Turn both blocks to the percent face: 25% + 25% = 50%! The physical reality never changes.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Front: 1/4 | Right: 0.25 | Back: 25% | Left: 90° (Geometric Equivalence)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/2], [1/4], [1/5], [1/10] + Desk Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 4–6

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 2/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE MULTI-NOTATION EQUATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you build a tower that shows '1/2' + '0.25' + '25%' and prove it totals 100% and 360°?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Stack [1/2] (Front) + [1/4] (Right: 0.25) + [1/4] (Back: 25%).**
   Expected Physical Finding: **Total height = 84.00 mm (1 Whole)**

2. **Sum their angles: 180° + 90° + 90°.**
   Expected Physical Finding: **360° (Full Circle)**

3. **Convert 3/4 to decimal on SC-6: [3] [ENTER] [4] [/].**
   Expected Physical Finding: **0.75**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [/] [4] [F-D] -> 0.25

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1	X: 1	Numerator
4	Y: 1 | X: 4	Denominator
/	X: 0.25	Decimal representation
F-D	X: 1/4	Toggle fraction / decimal display

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 4–6

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 4: Adding Like Fractions

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why didn't the size of the 1/8 blocks change when we stacked them?'
- Ask: 'What does the numerator tell you to do with your hands?'
- Ask: 'If you have 6/8 of a pizza, what simpler block matches that exact amount?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students writing 1/8 + 3/8 = 4/16. Hand them a 1/16 block and ask if their tower shrank!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a tower of like fractions with denominator 12. I level exactly with Block [3/4]. How many 1/12 blocks am I?"
""

OFFICIAL SOLUTION KEY:

Nine [1/12] blocks (9/12 = 3/4).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 1: Adding Like Fractions

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why don't we add the denominators when adding fractions? Building towers of eighths gives you an instant physical proof that the denominator never changes.

STEP-BY-STEP APPARATUS SETUP:

1. Pull four [1/8] blocks from your kit.
2. Stack three [1/8] blocks together (Tower 3/8).
3. Add one more [1/8] block to your tower (Tower 4/8).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks eight [1/8]s, four [1/4]s, two [1/2]s + Datum Stage

DATUM TARGET: 1/8 (10.5mm) × 4 = 42.00 mm = H(1/2)

CALCULATOR KEY: [1] [.] [8] [ENTER] [3] [.] [8] [+] -> 4/8 = 1/2

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 3B, Chapter 11 (Fraction Addition)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 4, Unit 2 (Addition of Fractions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 4, Ch. 7 • CCSS.MATH.4.NF.B.3.A • TEKS 4.3E
GRADE LEVEL BAND:	Grades 3–5 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 108–114
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 5 (Tutorial 1) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
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CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 2: LAB INVESTIGATION: THE EIGHTHS TOWER

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Stand your four [1/8] blocks beside Block [1/2]. Notice they level perfectly at 42.00 mm.

Step 2: If we added denominators, 1/8 + 3/8 would equal 4/16 (2.6 mm tall). Show that 4/16 would be tiny!

Step 3: Conclude: The denominator names the SIZE of the block (10.5 mm). The numerator counts HOW MANY blocks you hold.

Step 4: Test: Stack two [1/8] blocks. Verify they level with Block [1/4].

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
1/8 (10.5mm) × 4 = 42.00 mm = H(1/2)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks eight [1/8]s, four [1/4]s, two [1/2]s + Datum Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

STUDENT LAB SPREAD • LEVEL 1 & LEVEL 2 • Grades 3–5

STACKCALC ACCREDITED STEM INSTRUMENT

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 3/6 | INTERMEDIATE | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: BENCHMARK LEVELING

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

How many eighths must you add to 5/8 to reach the 84.00 mm whole summit?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Build Tower 5/8 (five [1/8] blocks). Measure remaining gap to whole.**
   Expected Physical Finding: Gap = 31.50 mm
2. **How many [1/8] blocks fit in 31.50 mm?**
   Expected Physical Finding: Three [1/8] blocks (3/8)
3. **Verify: 5/8 + 3/8 = 8/8 = 1 Whole.**
   Expected Physical Finding: Summit levels to 84.00 mm!

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [.] [8] [ENTER] [3] [.] [8] [+] -> 4/8 = 1/2

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1.8	X: 1/8	Key fraction 1/8
ENTER	Y: 1/8 | X: 1/8	Stack lift
3.8	Y: 1/8 | X: 3/8	Key fraction 3/8
+	X: 4/8	Add: 4/8 simplifies to 1/2 (0.5)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

STUDENT LAB SPREAD • LEVEL 3 & RPN BRIDGE • Grades 3–5

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Unlike Denominators

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did 12 work as a common denominator for both 3 and 4?'
- Ask: 'What is the connection between this lab and the Prime Studio LCM lab?'
- Ask: 'Could we have used 24ths to solve this problem? Why is 12 simpler?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding numerators and denominators (1/3 + 1/4 = 2/7). Show them that 2/7 would be shorter than 1/3 al
one!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve 2/3 + 1/6 using blocks. What is the sum, and does it simplify to a landmark block?"
""

OFFICIAL SOLUTION KEY:

2/3 = 4/6. 4/6 + 1/6 = 5/6 (Height = 70.00 mm).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Unlike Denominators

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

You cannot snap thirds and fourths together and read the answer on the blocks—unless you translate them into a common language: twelfths.

STEP-BY-STEP APPARATUS SETUP:

1. Stack one Block [1/3] on top of one Block [1/4] in the left comparator well.
2. Notice: The tower summit sits at 49.00 mm. No single block in your kit says '1/7' or '2/7'.
3. Challenge: How can we measure this tower using only identical pieces?

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks [1/3], [1/4], and seven [1/12]s + Twin Comparator Well

DATUM TARGET: H(1/3) + H(1/4) = 28.00 + 21.00 = 49.00 mm = 7 × H(1/12)

CALCULATOR KEY: [1] [.] [3] [ENTER] [1] [.] [4] [+] -> 7/12

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 5A, Chapter 3 (Adding Unlike Fractions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 2 (Fraction Equivalence & Operations)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 5, Ch. 2 • CCSS.MATH.5.NF.A.1 • TEKS 5.3H
GRADE LEVEL BAND:	Grades 5–6 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 114–120
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 2 (Leveling Datum) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE TWELFTHS TRANSLATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Replace Block [1/3] with [1/12] blocks. Notice: Exactly four [1/12] blocks level to 1/3 (4/12).

Step 2: Replace Block [1/4] with [1/12] blocks. Notice: Exactly three [1/12] blocks level to 1/4 (3/12).

Step 3: Combine both twelfths piles: 4 twelfths + 3 twelfths = 7 twelfths (7/12).

Step 4: Stand your 7/12 tower beside the [1/3]+[1/4] tower. Confirm: Both summits level perfectly at 49.00 mm!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
H(1/3) + H(1/4) = 28.00 + 21.00 = 49.00 mm = 7 × H(1/12)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks [1/3], [1/4], and seven [1/12]s + Twin Compa
rator Well

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: SUBTRACTION WITH UNLIKE BLOCKS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 1/2 − 1/3 physically. Place Block [1/2] and Block [1/3] side-by-side in the comparator stage.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure the height difference between 1/2 (42.0mm) and 1/3 (28.0mm).**
   Expected Physical Finding: **Difference = 14.00 mm**

2. **Which block fits the difference exactly?**
   Expected Physical Finding: **Block [1/6] (14.00 mm) or two [1/12]s**

3. **Write the equation: 1/2 − 1/3 = ?**
   Expected Physical Finding: **3/6 − 2/6 = 1/6**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [1] [.] [3] [ENTER] [1] [.] [4] [+] –> 7/12

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
1.3	X: 1/3	Key fraction 1/3
ENTER	Y: 1/3 | X: 1/3	Stack lift
1.4	Y: 1/3 | X: 1/4	Key fraction 1/4
+	X: 7/12	Evaluate sum with common denominator 12

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: Improper Fractions & Mixed Numbers

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 4/4 called a whole while 5/4 is called improper?'
- Ask: 'How does the 84.00 mm frame show the whole-number part of a mixed number?'
- Ask: 'Can you have a fraction greater than 2? What would 9/4 look like?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking 5/4 is an invalid number. Explain that 'improper' is just a traditional name for a fraction greater than 1.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am an improper fraction with denominator 6. I am equal to 2 1/2. How many 1/6 blocks make me up?"
""

OFFICIAL SOLUTION KEY:

2 Wholes = 12/6. 1/2 = 3/6. Total = 15/6 (fifteen [1/6] blocks).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/fraction-studio/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: Improper Fractions & Mixed Numbers

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What happens when you have more slices than fit in a single pie? In this lab, your towers break through the 84.00 mm ceiling into mixed numbers.

STEP-BY-STEP APPARATUS SETUP:

1. Stand the Whole Frame [1/1] (84.00 mm) on your desk.
2. Stack five [1/4] blocks in a single tower.
3. Notice: Your tower extends above the 84.00 mm frame!

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Fraction Blocks five [1/4]s, two [1/1] wholes, and [1/2] blocks

DATUM TARGET: 5/4 = 5 × 21.00 = 105.00 mm = 84.00 mm (1 Whole) + 21.00 mm (1/4)

CALCULATOR KEY: [5] [/] [4] [a b/c] -> 1 1/4

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 4A, Chapter 5 (Improper Fractions)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 4, Unit 4 (Fractions Greater Than 1)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Grade 5, Ch. 2 • CCSS.MATH.4.NF.B.3.B • TEKS 4.3B

GRADE LEVEL BAND: Grades 4–6 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 120–126

HARDWARE UNBOXING BOOKLET: FRAC-01 Booklet: Page 4 (Fill Datum) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: BREAKING THE CEILING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Measure how much your five [1/4] tower extends above the 84.00 mm whole frame: exactly 21.00 mm.

Step 2: Stand a loose [1/4] block on top of the whole frame. Notice: It levels with your tower!

Step 3: Record: 5/4 is the exact same physical amount as 1 Whole + 1/4 (1 1/4).

Step 4: Now stack seven [1/3] blocks. How many whole frames does it fill? (2 Wholes + 1/3 = 2 1/3 = 7/3).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
5/4 = 5 × 21.00 = 105.00 mm = 84.00 mm (1 Whole) + 21.00 mm (1/4)

PHYSICAL APPARATUS MANIFEST:

Fraction Blocks five [1/4]s, two [1/1] wholes, and [1/2] blocks

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: DECOMPOSING IMPROPER FRACTIONS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Can you convert 11/8 into a mixed number using only blocks, and verify it on your calculator?

FORM A MASTER STATION TASKS (Complete All 3):

1. **Stack eleven [1/8] blocks. Compare against the 84.00 mm whole.**
   Expected Physical Finding: **Tower height = 115.50 mm**

2. **How many whole frames are filled, and what is the remainder?**
   Expected Physical Finding: **1 Whole (8/8) + 3/8 remainder**

3. **Key [11] [ENTER] [8] [/] on SC-6.**
   Expected Physical Finding: **1.375 = 1 3/8**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [5] [/] [4] [a b/c] -> 1 1/4

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
5	X: 5	Numerator 5
4	Y: 5 | X: 4	Denominator 4
/	X: 1.25	Decimal evaluation
a b/c	X: 1_1/4	Convert to mixed number 1 and 1/4

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Rational Stacks & RPN Operations

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did dividing 3/4 by 1/8 give a whole number answer of 6?'
- Ask: 'How is division with fractions just asking: how many times does block B fit in block A?'
- Ask: 'How does the RPN stack keep track of multi-step fraction problems?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to invert and multiply without understanding why. Ground them in the physical question: 'How m
any small blocks fit in the big block?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Solve (1/2 + 1/3) ÷ 5/12 on the RPN stack. What is the final answer?"
""

OFFICIAL SOLUTION KEY:

1/2 + 1/3 = 5/6 = 10/12. (10/12) ÷ (5/12) = 2.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

FACILITATOR / TEACHER SIDE • FRAC-01 Booklet: Page 7 (RPN Stack Mapping) & Page 8 (Exercises)

CALIFORNIA DESIGN LAB • VERIFIED REPRODUCIBLE

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Rational Stacks & RPN Operations

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

When you solve fraction arithmetic on paper, you must remember flip-and-multiply rules. In the RPN calculator, operations execute exactly like stacking and unsta

STEP-BY-STEP APPARATUS SETUP:

1. Place Block [3/4] on your desk. On top of it, place Block [2/3].
2. Key [3] [.] [4] into your SC-6 or SC-32 calculator. Press [ENTER].
3. Key [2] [.] [3] into Register X. Press [*].

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Full Fraction Kit + SC-6 Elementary / SC-32 Scientific Calculator

DATUM TARGET: Direct Isomorphism: Top of Physical Tower = Register X, Bottom = Register Y

CALCULATOR KEY: [3] [.] [4] [ENTER] [2] [.] [3] [*] -> 1/2

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 6A, Chapter 3 (Fractions & Decimals Operations)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 5 (Multiplying & Dividing Fractions)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 6, Ch. 2 • CCSS.MATH.6.NS.A.1 • TEKS 6.3B
GRADE LEVEL BAND:	Grades 5–8 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 1: SC-6 Elementary Companion, Ch. 5, pp. 126–134
HARDWARE UNBOXING BOOKLET:	FRAC-01 Booklet: Page 7 (RPN Stack Mapping) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

STUDENT ENTRANCE SIDE • Fraction Studio Proportional Towers • Grades 5–8

CALIFORNIA LAB QC SPECIFICATION // STACKCALC

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: FRACTION MULTIPLICATION AS SCALING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Look at the result on your calculator LCD: 1/2 (or 0.50).

Step 2: Stand Block [1/2] beside your desk. Why is 2/3 of 3/4 equal to 1/2?

Step 3: Three [1/4] blocks divided into three groups gives 1/4 per group. Two of those groups equal two [1/4]s = 1/2!

Step 4: The physical blocks confirm the RPN stack calculation: scaling 3/4 by 2/3 shrinks it to exactly 1/2.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Direct Isomorphism: Top of Physical Tower = Register X, Bottom = Register Y

PHYSICAL APPARATUS MANIFEST:

Full Fraction Kit + SC-6 Elementary / SC-32 Scientific Calculator

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

FRAC-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: FRACTION DIVISION WITHOUT FLIPPING

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Model 3/4 ÷ 1/8 using blocks. How many [1/8] blocks fit inside Block [3/4]?

FORM A MASTER STATION TASKS (Complete All 3):

1 **Stack [1/8] blocks beside Block [3/4] until they level.**
Expected Physical Finding: **Six [1/8] blocks level to 3/4**

2 **Execute on SC-6: [3] [.] [4] [ENTER] [1] [.] [8] [/].**
Expected Physical Finding: **Result = 6 (Exact integer!)**

3 **Explain why 3/4 ÷ 1/8 = 6 without using paper rules.**
Expected Physical Finding: **Six eighths fit inside three fourths.**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [3] [.] [4] [ENTER] [2] [.] [3] [*] -> 1/2

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
3.4	X: 3/4	Key fraction 3/4
ENTER	Y: 3/4 | X: 3/4	Stack lift: 3/4 moves to Y
2.3	Y: 3/4 | X: 2/3	Key fraction 2/3 into X
*	X: 1/2	Multiply: (3/4) × (2/3) = 6/12 = 1/2

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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